

CYC Standard Data Format

What is a CYC file?

A CYC (Cycle Counting) file contains the counted fatigue cycles obtained from an LDS file using a selected cycle counting method.

File naming convention

File: CYC_{ResearcherLastname}_{YYYY-MM}_{method}.csv

- YYYY-MM = starting month of the experiment
- method = range_mean, range_pair, simplified_rainflow, etc.
- Example: CYC_Dupont_2024-03_range_mean.csv

File format requirements

- Encoding: UTF-8
- Separator: comma (,)
- File type: .csv
- Header required
- Units: MPa

Required columns (column names must be exact)

Variable name	Description	Symbol	Unit	Data type	Mandatory
stress_range	Stress range ($= 2 * \sigma_a$)	delta_sigma	[MPa]	double	y
stress_mean	Mean stress	sigma_mean	[MPa]	double	y
stress_ratio	Stress ratio	R	[-]	double	y
n_cycles	Number of cycles	n		double	y
cum_n_cycles	cumulative number of cycles for a given bin of σ_a , σ_{mean} and R.			double	y

Example of valid CYC file

stress_max
-2
1
-3
5
-1
3
-4
4
-2

LDS file



stress_range	stress_mean	stress_ratio	n_cycles	cum_n_cycles
3.0390625	-0.5078125	-2.003861004	0.5	87.5
3.9765625	-0.9765625	-2.930501931	0.5	75
3.9765625	0.9921875	-0.334207077	0.5	62.5
6.0078125	0.9921875	-0.50342131	0.5	50
6.0078125	1.9765625	-0.20627451	0.5	37.5
7.0234375	-0.5078125	-1.33810143	0.5	25
7.9609375	0.0078125	-0.996082272	0.5	12.5
7.9609375	0.9921875	-0.600942655	0.5	0

CYC file